# A Hybrid Annotation Model for Arabic Argumentative Debate Corpus

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**Abstract.** This study introduces the first hybrid annotation model for Arabic argumentation debate corpus. It aims to analyze Arabic argumentation structure in this competitive debate corpus using this hybrid model. The model combines Aristotle's three appeals of logos, ethos and pathos with Toulmin's model of argument structure analysis, in addition to some added labels inspired by the Arabic debate corpus. These added labels are self- repetition, team repetition and reciting, which were added to the hybrid model to reflect the uniqueness of the Arabic argumentation dataset. In addition, the hybrid model further subdivides Toulmin's 'backing' into evidential backing and rational backing reflecting patterns emerging from the dataset annotation. The study presents preliminary findings, and significant patterns of the used labels in Arabic argumentation. The model addresses a gap and an under-representation of Non-English language argument annotation models by enhancing linguistic diversity in argument structure analysis, argument mining and Natural Language Processing (NLP). Thus, the model eliminates Artificial Intelligence (AI) bias by supporting a low-resource language such as Arabic. The multifaceted nature of the corpus on which the model was used, in terms of topics, gender representation, and geographic origins of debaters, serves as a robust resource for validating the model and for analyzing argumentative discourse in the Arabic speaking world. The implications of applying this hybrid model in Arabic argumentation research will be discussed.

**Keywords:** Argumentation · Annotation · Toulmin's model

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## 1 Introduction

Aristotle, in his On Rhetoric, particularly the second chapter of this important treatise argues that persuasion can be achieved in three ways. One is the truthfulness and logicality of the argument, later known as logos. Two is the speaker's ability to get the audience to trust him. Three is the set of emotions the speaker is able to instill in the audience to make it accept his point of view. In modern rhetoric, these three ways have been named logos, ethos and pathos respectively [1]. Stephen Toulmin (1922–2009), who is "one of the founding fathers of modern theory of argumentation" [2], proposed a model that inherited the key features of classical rhetoric [3]. Toulmin's aim was to get argumentation out of the "abstract and formal criteria of mathematical logic" [4,5], and propose a model that takes argumentation as a human endeavor incorporating methods that we "actually use in everyday life" [4]. In this study, we present a hybrid model of argument annotation for Arabic competitive debates mainly based on Aristotle's and Toulmin's models.

Competitive debating, is a rigorous intellectual activity involving oral argumentation. It is highly regulated and is commonly held in debating contests and tournaments. University and school students of both genders from all over the world participate in Arabic debating tournaments on the local and international levels. QatarDebate Center (www.qatardebate.org) is the Arab leading debate institution, promoting debating culture in the Arab world and organizing major debating events. QatarDebate's 3 vs 3 debate format as shown in Fig. 1, which is a customized format of the World Schools Debating Championship (www.wsdcdebating.org), is the adopted format in Arabic competitive debating activities. In this format, two teams debate a motion; each team opposes the other with one representing the 'proposition' and the other the 'opposition'. Each team is composed of three speakers, and each speaker is given a total of 6–7 min to present their arguments. The process begins with the first proposition speaker, and then the first opposition speaker, and so on until the last speaker from the opposition. Then, the Reply Speech which is a 3-min technical speech delivered by each team to conclude the debate. A debate is typically adjudicated by a panel of odd number judges who select the winner based on rigoros rubrics and descriptors that help in designating the right scores for each of the two teams. The quality of the arguments presented and the rebuttal provided are the main focus of the panel to adjudicate the debate.

This study introduces an annotated corpus of Arabic competitive debates. For the implementation and validation of the model, a corpus of Arabic argumentative debate is used. Currently, it stands as the most extensive, diverse and specialized corpus of Arabic argumentative content. The uniqueness of the model originates from the nature of the competitive debate which is rich in structured argumentative content and explicit sentiment inherent in the debating enterprise. The data exploration provides an empirical basis for the hybrid model the authors propose. The expanded model will inform Arabic rhetorical and linguistic studies of spoken Modern Standard Arabic (MSA) argumentation, and will assist in developing Arabic NLP tools for argument mining. This

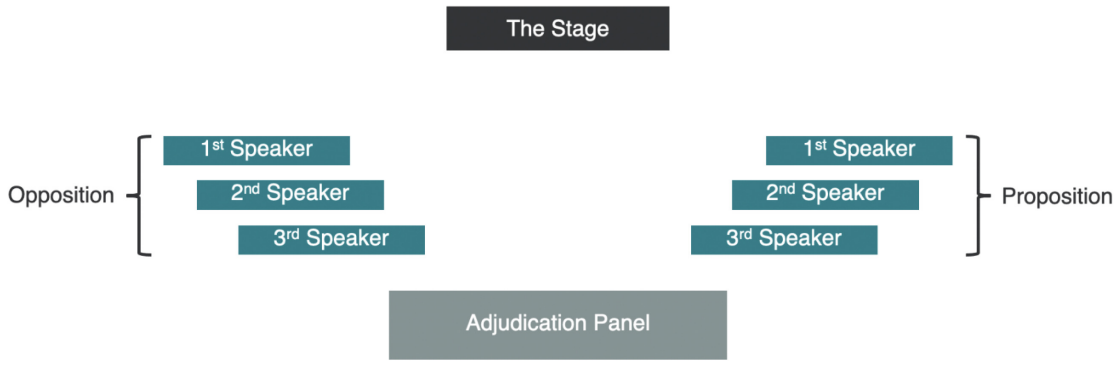


**Fig. 1.** Illustration of 3 vs 3 Debate Format

unique Arabic annotated corpus of competitive debates can serve as primary training material for Arabic argument mining as well as discourse analysis of Arabic argumentation strategies.

## 2 Theoretical Framework

### 2.1 Argument Analysis

Two key models of argumentation analysis are commonly referred to in the literature. The first is Aristotle's model in which he proposed the three key terms logos, ethos and pathos which are used in many argument analyses in different languages. Logos refers to the content of the speech or oration, ethos refers to the credibility of the speaker, and pathos refers to how the audience is emotionally engaged in the speech or oration [1]. The second model is Toulmin's model [6] in which he proposes six main elements of an argument: Claim (C) which is the conclusion we aim to make. This is the statement the arguer makes and needs to justify and support. The support comes in the form of Data (D), later called Grounds. These are the facts we treat as the premise of our claims or arguments. These data might be challenged in the course of an argument, so they need to be supported by a Warrant (W), a proposition showing the logical link between the data and the claim. Toulmin [6] then talks about another element of argument skeleton, which he calls Qualifier (Q), which refers to the scope of certainty or otherwise of the claim. Toulmin then suggests another element in the argument structure and calls it Rebuttal (R) showing the refutation of the claim and the application of a qualifier to produce a more restricted or qualified claim. The last element of Toulmin's model is Backing (B), which is the support that adds authority and legitimacy to the claim, the warrant and the data.

### 2.2 Related Studies

As research in argumentation analysis and argument mining relies heavily on annotated corpora of argumentative texts, we identify some relevant efforts on corpus creation for argument mining, such as the work of [7]. In this work, the authors describe the creation of the Argument Annotated Essays Corpus containing 302 persuasive student essays annotated for argument components derived from Toulmin's model. The corpus supports the investigation of argument structure in free-text essay writing. Furthermore, [8–10] developed the Qatari Corpus of Argumentative Writing of approximately 200,000 tokens of Arabic and English writing by undergraduate students. The corpus utilized part-of-speech (POS) annotation with TreeTagger for English texts and Farasa for Arabic texts. The students are native Arabic speakers and fluent in English. The team asked the students to write an Arabic and English essay based on specific argumentative prompts and instructed them to include in their essays a clear thesis statement supported by relevant evidence. These studies demonstrate growing efforts to create corpora annotated with argumentation elements to support computational modeling. However, the Qatari Corpus annotation focuses on micro-level categories of argumentative essay writing, hence its purely pedagogical orientation. To the best of our knowledge, the hybrid model proposed in our study is the first model that targets macro-level argument structure in a debate corpus annotating spoken arguments as discourse elements, hence the uniqueness and originality of our hybrid model.

Some studies have proposed various annotation methodologies for languages other than Arabic. For example, [11] conducted an annotation analysis of argument schemes such as expert opinion following the *Argumentum Model of Topics*. [12] annotated claims and evidence in controversial topics for automatic detection in a simple claim/evidence model. There are argument mining efforts in other languages such as Italian [13] and the argument relation annotations from multilingual social media such as X (Twitter, formerly), [14]. [15] presented the UKP Sentential Argument Mining Corpus which annotates 25,492 sentences from online comments according to a simplified model with claims, premises, and major claim tags. The corpus focuses on Components at the sentence level within user-generated web discourse.

Aristotle's rhetorical argumentation model, although has been a source of inspiration for rhetorical studies for centuries, has been critiqued by a number of scholars, including Toulmin. In relation to Aristotle's universal orientation, Toulmin [6] points out that Aristotle was an idealist as he was "much concerned with syllogisms in which both the premises were universal" (p. 100) as opposed to Toulmin's interest in propositions that are used to "justify particular conclusions about individuals" (ibid). The work in [16] describes Aristotle's Rhetoric as "difficult to read, full of discrepancies, gaps and repetitions" (p. 128). Quandahl observes that the model needs to be adapted with caution since it was developed in an entirely different historical context. Kennedy also voices similar concerns in [1]

Similarly, Toulmin's argumentation model has been used in text linguistics, discourse analysis, argument mining, rhetoric, and writing pedagogy [4,7,17–26]. It has not been without drawbacks, however [27,28]. In the numerous applications of Toulmin's model, a number of issues have been raised about the model. For example, some scholars comment on its ambiguity, inconsistency and vagueness [29]. Others describe it as rigid ( [19] and needs "to be flexible - not tied to one standard function of argument" (p, 3). [7] highlight "the need for greater efforts in building a framework in which argument mining tasks are carried out, covering all aspects of agreement on the argument theoretical concepts being identified" (p. 806). Toulmin's model is also described as idealistic and gets even frustrating when used to analyze real life data [30]. Scholars have called for a modification or expansion of the model [31].

[25] examine the application of Toulmin's model of argumentation in Chinese EFL argumentative writing. They claim that not all the elements "are explicit, and sometimes warrants may not need to be stated in real life arguments" (p. 445). [32] comments on Toulmin's model referring to the element of warrants. He contends that Toulmin's model is "problematic at points, especially his discussion of warrants, in a way which leads to a confusing misinterpretation of how his model should be understood" (p. xii). He rejects "Toulmin's distinction of data, warrants and backing as appropriate for analyzing arguments as products" (ibid). He devotes a whole chapter to discussing Toulmin's "problematic notion of warrant" (cf. chapter 3). Similarly, Toulmin's notion of data being restricted to 'facts' has been questioned [33]. Toulmin's notion of 'data', according to Herman needs to be expanded to cover other elements such as opinions and interpretations, and Toulmin's notion of 'backing', needs to be expanded to include backing for data and backing for warrant [33].

Despite these comments, Aristotle's and Toulmin's models have been widely used to label argument structure in various linguistic contexts. To accommodate these different contexts, it has been modified and expanded to accommodate various types of data from different genres and language situations. [20] modify Toulmin's model to make the components fit with the purpose of argument analysis of students' writing. Their modified model includes final claim, primary claim, counterclaim, rebuttal, data and other (p. 174). [22] reduce Toulmin's six elements to two "the main components of every argument, and the rest are supporting sub-arguments that may or may not exist in an argument." (p. 5). [24] use a modified model of Toulmin's which is similar to that adapted by [20] referred to above, except in the last category which is 'other' in [20] but 'supporting reason or example' in [24], (p. 160).

Both Aristotle's and Toulmin's models of argument analysis are significant and important in core argumentation theory and analysis because they assist researchers in analyzing and labelling the elements that make an argument in discourse. As shown from the review above, the models seem to have a wide scope of applicability in various areas of inquiry. The current paper contributes with a hybrid argumentation annotation model for Arabic debate used on an original Arabic debate corpus. Both the model and the corpus can serve as

a benchmark for argumentation mining methods in the Arabic language. The speeches in the corpus come from competitive debates, which are structured and formal forms of argumentation. The debaters' backgrounds are demographically diverse, resulting in more representation of argumentation among native Arabic speakers. One of the preliminary studies [34] examines using this annotated corpus to classify the modes of persuasion in these debates and considers the evaluation of language models in the classification task with respect to speakers' gender to study the fairness in these language models.

### 2.3 The Proposed Hybrid Model

The hybrid model proposed in this study is inspired by Aristotle's model of argumentation which sees argumentation as a rhetorical experience involving three main types of argumentative appeals: logos, ethos and pathos [1] and is supported by the various modern applications of these three classical appeals in various types of communication and discourse analysis [35–41]. Another source of support for our hybrid model comes from [42] who argue that the warrant in Toulmin's model that binds both the data and the claim can come in three different ways. These are (1) "relations existing among phenomena in the external world" (p, 48), (2) "an assumption concerning the quality of the source from which the data are derived" (ibid) and (3) "an assumption concerning the inner drives, values or aspirations which impel the behavior of those persons to whom the argument is addressed" (ibid). We use Toulmin's model to bridge the gap of detail in Aristotle's model and we situate Toulmin's model within the Aristotelian logos.

## 3 Methods

As shown in Table 1, we take (1) in [42] to mean syllogistic argumentation, which is the component directed to the content of the syllogism (rational and dialectical argumentation) and this corresponds to Aristotle's logos and to Toulmin's claim, grounds, warrant, backing, qualifier and rebuttal. We further subdivide Toulmin's 'backing' into two types: rational backing and evidential backing to provide more granulated analysis of the annotated corpus in response to patterns emerging from the data. Moreover, to respond to insights from the data we also added two more elements. The first one is repetition for which we identified two further sub-elements: self-repetition where the speaker repeats himself/herself throughout his/her speech, and team-repetition where the speaker repeats claims, positions, or rebuttals mentioned by his/her team earlier. The second element is 'reciting' where the speaker repeats the argument of the opposing team to refute it. We take (2) in [42] to mean the expressive argumentation, which relates to the speaker's appeal to create their own credibility and trustworthiness, which directly corresponds to Aristotle's ethos. Therefore, we added ethos to the model to account for this kind of expressiveness in argumentation, where the speaker's focus is on expressing, maintaining and preserving his/her

**Table 1.** The hybrid annotation model based on Aristotle's and Toulmin's models with added labels

| No | Argumentation Type |
|---|---|
| 1 | **Inferential Argumentation (Logos)**<br>Definition: Argumentation based on the logical structure of an argument. This is Aristotle's Logos and is substantiated by the categories proposed by Toulmin with some more labels added in response to the debating corpus of this study<br>(1) Position: Team's position in the debate i.e. either proposition or opposition<br>(2) Super claim: An assertive statement made by a debating team which outlines the team's overall goal<br>(3) Claim: An assertive statement or a conclusion that the proposition team seeks to justify and support and the opposition team seeks to refute and rebut<br>(4) Grounds: Usually agreed upon information and facts that support the claim. These include definitions<br>(5) Warrant: The logical link that connects Ground to the Claim - often implicit<br>(6) Rational backing: Logical and inferential reasoning including justifications, conclusions and mini supported assertions<br>(7) Evidential backing: Specific examples or research findings provided to support a claim<br>(8) Qualifier: A statement representing a concession on the part of the speaker as to the scope or certainty of the claim, whether in response to a rebuttal or not<br>(9) Rebuttal: The refutation of a claim by an opposing team or individual<br>(10) Self-repetition: Partial or total repetition of statements made by the same speaker<br>(11) Team repetition: Partial or total repetition of statements made by the team of the speaker<br>(12) Reciting: Partial or total repetition of statements made by the opposing team |
| 2 | **Expressive Argumentation (Ethos)**<br>Definition: Argumentation based on the expressive function of discourse whereby a speaker uses statements that help to create their own credibility, morality, trustworthiness, organization, and order. This type of argumentation includes: authority, honesty, trustworthiness, reputation, fairness, transparency, openness, accessibility, etc. |
| 3 | **Appellative Argumentation (Pathos)**<br>Definition: Argumentation based on the appellative function of discourse whereby a speaker uses statements that appeal to audience emotions and feelings and attempt to engage the receivers in one way or another in the discourse. This type of argumentation includes: empathy, fear, anger, love, compassion, hope, patriotism, uncertainty, etc. |

own credibility, moral and ethical standing as a way of persuading the audience to win their trust. Finally, we take (3) in [42] to mean the appellative argumentation, which appeals to audience's feelings and emotions, and attempt to engage the audience in one way or another in the discourse. Thus, we integrated the element of 'pathos' from Aristotle's model to account for this appellative dimension of argumentative discourse. This hybrid model is theoretically grounded in the work of both Aristotle and Toulmin and is empirically grounded in the patterns of data emerging from the annotation process that reflect the nature and uniqueness of Arabic debating argumentation in terms of being spoken and displaying various patterns of repetition in presentation and argumentation. The hybrid model with its new integrated labels would serve as an annotation model for various forms of Arabic debating argumentation. Once, it is finally validated against further data annotation and analysis (the annotation of the entire corpus

120 debates), the model can be tested on Arabic argumentation analysis outside the debating context.

### 3.1 Corpus Building

In this argument annotation study, we used 40 debate transcripts from Munazarat 1.0 corpus. The corpus has 204,294 word tokens and 21,692 word types. The details of the full corpus are described in [43]. All debates in this corpus comply with the following criteria: (1) follow QatarDebate Center format (3 vs 3) illustrated in Fig. 1 above and explained in more detail in [44], (2) are held in a physical (face-to-face) debate setup, (3) involve university students and native speakers of Arabic. Tables 2, 3, and 4 show some of the important variables of this corpus. We maintain a balanced male-to-female ratio in the corpus as shown in Table 2. The corpus represents debaters from 38 universities that represent 13 Arab countries as shown in Table 3. Also, a variety of debated topics are represented in the corpus (11 topics) as shown in Table 4.

**Table 2.** Gender Representation

| Gender | No of Speeches |
|---|---|
| Male | 126 |
| Female | 114 |
| Male/Female Ratio | 1.08 |

**Table 3.** Country Representation

| Country | No of Debating Teams |
|---|---|
| Qatar | 14 |
| Jordan | 12 |
| Sudan | 11 |
| Oman | 10 |
| Tunisia | 9 |
| Kuwait | 6 |
| Lebanon | 5 |
| Libya | 5 |
| Palestine | 3 |
| Bahrain | 2 |
| Iraq | 1 |
| Algeria | 1 |
| Somalia | 1 |
| Total | 80 |

### 3.2 Corpus Annotation

For the annotation of the corpus, we take every discourse segment in the debate as a piece of argumentation following [45][p. 172] who argues that 'whenever there is persuasion, there is rhetoric. And wherever there is meaning, there is persuasion.' Therefore, we analyze every piece of the debate and treat every text in the corpus as example of argumentation. Some studies classify their corpora into argumentative vs non-argumentative discourse [46]. However, from a critical discourse analysis perspective [47,48], we believe this separating approach is limited and is probably motivated by a reductionist representational view of language. Here we believe that discourse in general is socially constitutive and motivated [48,49]. The annotated corpus so far completed, 40 annotated debates, represents three variables: gender, country and topic as shown in Tables 2, 3 and 4.

**Table 4.** Topic Representation

| Topic | No of Debates |
|---|---|
| Politics | 11 |
| Human rights | 8 |
| Ethics/Philosophy | 7 |
| Sports | 3 |
| Technology | 3 |
| Environment | 2 |
| Culture | 2 |
| Law | 1 |
| Lifestyle | 1 |
| Education | 1 |
| Media | 1 |
| Total | 40 |

### 3.3 Annotation Model

In Table 1, we describe the various elements in our hybrid model (Aristotle, Toulmin + our added labels). The model provides the type of argumentation along with the labels used in each type of argumentation. It also provides a definition for the types and labels used in the annotation process. The labels outlined in Table 1 were used as coding labels for argument structure elements in the corpus. We also provide a set of linguistic triggers that help the annotators to smoothly identify and label the argument segments.

### 3.4 Annotation Process

In this subsection, we describe the annotation process providing details of the annotation model used, the annotation team and their training, the inter-annotator agreement and briefly outline the challenges encountered.

#### 3.4.1 Annotation Tool

The annotation process is conducted using a paid digital tool called "UBIAI" (https://ubiai.tools/) which is a website that allows for a team of annotators to separately make text labeling on the same text file in a user-friendly environment. Each debate is annotated by analyzing its text using the labels identified in the hybrid model proposed in this study under 14 different labels as shown in Table 1.

#### 3.4.2 Annotation Team

The annotation team was carefully selected to ensure the best quality and high diversity amongst its members. It consisted of four native speakers of Arabic who are also experts in Arabic competitive debates. They have developed considerable expertise within QatarDebate Center as experienced debaters, debate judges, and debate trainers (3 males & 1 female). The annotators come from different academic backgrounds (Engineering, Foreign Affairs, and English Literature) to ensure in-depth knowledge of the annotators in the subject areas covered in the debates. In addition to the annotators' expertise in Arabic competitive debating as debaters, adjudicators, and trainers, they were also provided with intensive training to enhance their annotation and discourse analysis skills.

#### 3.4.3 Annotation Training

The annotator training process entailed the formulation of a structured annotation framework. While this framework is based on Aristotle's and Toulmin's models, custom-made annotation labels were judiciously integrated to reflect the insights emerging from the data as the annotation process evolved. As mentioned earlier, the data exploration in the annotation process provided an empirical foundation for the hybrid model which the authors propose in this study. Subsequent to model development, this hybrid model was thoroughly presented to and discussed with the annotators. Leveraging the annotators' experienced insights as active debaters, the framework underwent meticulous iterative enhancements, ensuring its alignment with the focal argumentation model. This iterative and collaborative refinement process culminated in the ultimate crystallization of a definitive annotation model, which served as the key guide for the annotators' meticulous work.

We used a double annotation process for the first twenty debates where two different annotators separately annotated each debate transcript. Then a reviewer, an academic who was part of the team developing the hybrid annotation model, would look at the IAA rate calculated by the UBIAI tool, read both

annotation files, compare the differences and edit the final annotated file. This process was repeated for every debate in the first set of 20 debates. Detailed feedback was given to annotators through a series of 3-hour weekly meetings with the reviewers. Despite the fact that this process was costly in terms of time, effort, and resources, it was necessary to ensure that all debates were being annotated in a consistent manner. The annotation procedure aimed to achieve as precise and descriptive annotation as possible. We did not address the quality, effectiveness, or integrity of the arguments because we did not intend to introduce any further distraction to our descriptive analysis of the corpus by involving evaluative judgments on the quality of the arguments.

#### 3.4.4 Inter-Annotator Agreement (IAA)

When we calculated the IAA for the first 20 debates, we noticed that the IAA ranged mostly between 20 and 50 for the first half of the debates annotated in the early stages. However, after the three weekly meetings between annotators and reviewers (researchers) to fine-tune and closely refine the annotation model, the IAA for the entire corpus increased to be in the range of 50 to 75, corresponding to 0.50–0.75 in Cohen Kappa's test and indicating substantial agreement that is comparable to other studies [7,50].

#### 3.4.5 Challenges Encountered

The argumentation annotation process cannot be as precise as parts-of-speech annotation, for example, due to the subjectivity involved in identifying and analyzing argument segments as discourse segments. For example, two annotators would agree on annotating one segment under the same label, yet they would disagree on the start or end of the segment by one or two words. In addition, one of the annotators might divide the same short paragraph into two or more parts of the same label while the other annotator would keep the whole short paragraph under one label. Thus, most disagreements were insignificant and acceptable as they did not represent contradictions between the annotators. Most disagreements were attributed to the nature of the studied speech corpus which allows for some kind of ambiguity in terms of argument segmentation.

## 4 Initial Findings and Discussion

In this section, we present some key initial findings from the 40 annotated debates. As shown in Table 5, the highest average word count in the dataset goes to 'rebuttal' 24.24%. Based on the definitions provided in the tool in Table 4, rebuttal is defined as the refutation of a claim by an opposing team or individual. Rebuttal is the second most important element in a debate after 'claim'. It is interesting, however, that 'claim' is less used than rebuttal, only 5.40% of the word count in the dataset. This is justified by the fact that a debate normally includes fewer claims than rebuttals, and a claim can be refuted in different

ways by more than one debater. In addition, the label 'rational backing' often includes mini-claims, while in the dataset, it was annotated using a different label which received 15.58% of the word count of the entire corpus, so this also justifies the relatively less frequent mention of 'claim' in the dataset compared to 'rebuttal'. This percentage faithfully reflects the desire of every debating team and every member of the team to support their claims with as much rationalization as possible. In our model, we separated 'rational backing' from 'evidential backing'. If taken together, both types of 'backing' would make 20% of the word count, which puts the label comfortably as the second highest label in terms of word count in the dataset after 'rebuttal'.

The label 'ethos' received 13.72% of the overall word count in the dataset which is a relatively high percentage of use in the corpus. Ethos has two functions: one is expressive, i.e. expressing speaker's credentials to create credibility and believability, and the second is organizational, providing discourse signposting to make it easy for the audience to follow the thread of the argument in the debate, which ultimately feeds into the credibility of the speaker as an organized and confident speaker. This high percentage justifies our hybrid model taking this element from Aristotle's model. In fact, the label of 'ehtos' is so common in our dataset that it cannot just be ignored. In competitive debating as in any other type of debating, the speaker's credibility and image are crucial in making the audience trust and believe the speaker, hence its importance in an argumentation model that seeks to analyze natural language in authentic argumentation settings. This finding echoes the importance of ethos in argumentation and communication in general [35–37,39–41].

Analyzing the distribution of labels based on the role of the speakers (first, second, third) shows the relevance of the argument labels proposed in the hybrid model to the role of the speaker. As shown in Fig. 2, The labels grounds, position, and super claim appear to be associated mainly with first speakers which is justified given their role in defining the terms of the motion and the stance of the team. Interestingly, the use of rational backing is absent in third speakers' argumentation because of their main role which is refuting the other team. The high use of rational backing compared to evidential backing across the three roles suggests a tendency of debaters to focus more on logical argumentation than using real-life examples which is a distinct feature of competitive debates. The lack of using warrants and qualifiers reflects a long standing problem with Tolmin's model which is the ambiguity of the concept of warrant and qualifier [3, 7,21,51–53]. The high use of rebuttal in the three roles suggests that the use of refutation is a very common argumentative component among debaters. Finally, the use of ethos and pathos across the three roles is consistent which suggests a general tendency to use those techniques in argumentation. It is interesting that the use of ethos is particularly higher compared to pathos, which indicates the tendency of debaters to focus more on establishing trust in the audience than relying on emotions.

For male and female, there is a high consistency in the distribution of labels across genders, which suggests that the gender of the speaker is not a factor

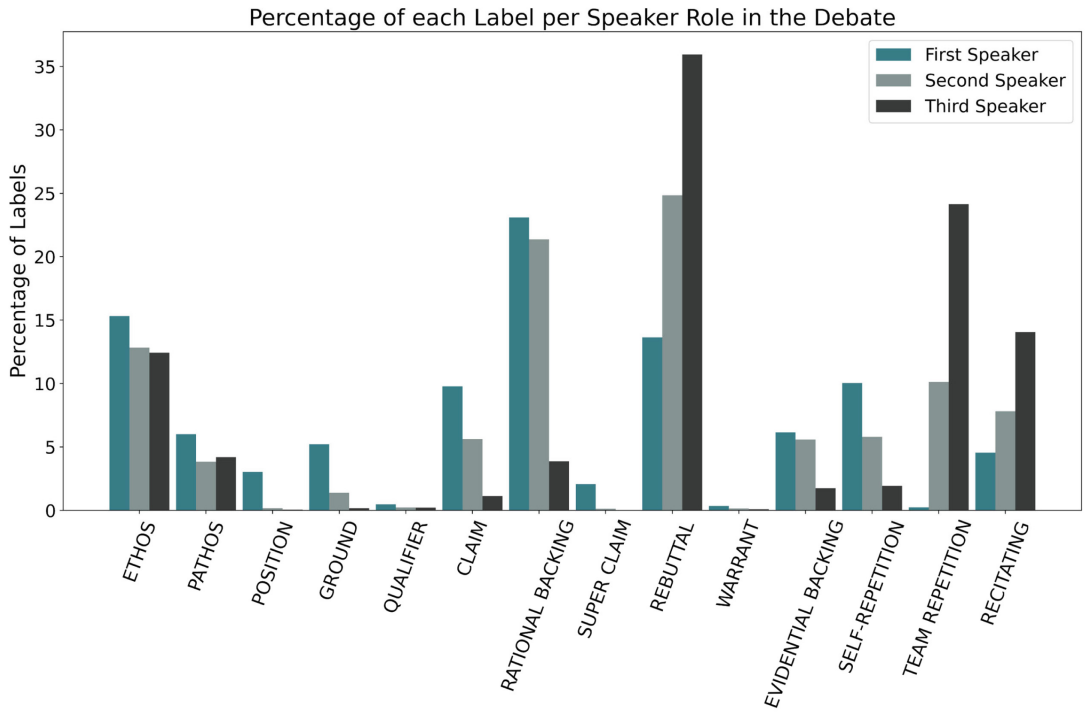


**Fig. 2.** Label distribution across debaters' roles

that affects the use of argumentation labels by debaters. This consistent pattern between both genders might be attributed to the well-defined structure of competitive debates compared to other forms of argumentation where the structures are relatively less well-defined.

The results also show that the elements added in our hybrid model, i.e. self-repetition, team repetition and reciting, if taken together, make the highest entity in terms of average word count in the entire dataset, 26.27%. This high percentage again explains why this crucial element had to be added to our hybrid model. It reflects the natural language used in spoken argumentative debates where repetition tends to serve an important function, i.e. to state, restate and reaffirm the claims, the backing or the grounds. In competitive debating particularly, debaters are expected to repeat their arguments, the arguments made by their teams and those made by their opponents. Attending to this key feature of debate argumentation in our hybrid model is a move that is supported by these figures which show how significant repetition is in Arabic argumentative debates. This finding echoes arguments by other Arabic argumentation scholars [49,54–59].

Another interesting result is the very low visibility of the argument labels 'warrant' and 'qualifier' in the annotated dataset, only 0.30 and 0.31 of the word count of the entire corpus respectively. This is not surprising because 'warrants' are not easy to identify and annotate. In fact, much of the discussion on Toulmin's model has been on this argument element in particular [3,7,21,51–53]. Similarly, the label 'qualifier' seems to have a very low visibility in the dataset.

However, while the reason behind the low visibility of 'warrants' in the data could be attributed to the difficulty the annotators face in locating it, the reason for the low visibility of 'qualifier' could be due to the reluctance of debaters to use it. Since it is a form of concession, it seems debaters tend to avoid it. However, this conclusion needs to be further examined when the entire corpus is fully annotated. The model we develop for Arabic debate annotation reflects the same issues with the vagueness of 'warrant' as expressed by these scholars [3,7,21,51–53]. The reason behind this vagueness is that it is often implicit and it is expressed as a logical principle or assumption and this implicitness makes it difficult for annotators to spot the warrant.

**Table 5.** Statistical distribution of argument labels in the dataset

| Label | Mean % | Median % | Standard Deviation |
|---|---|---|---|
| Ethos | 13.72 | 13.66 | 0.0321 |
| Pathos | 4.91 | 4.82 | 0.0200 |
| Position | 1.08 | 0.83 | 0.0063 |
| Grounds | 2.34 | 1.93 | 0.0140 |
| Qualifier | 0.31 | 0.20 | 0.0069 |
| Claim | 5.40 | 4.88 | 0.0255 |
| Rational Backing | 15.58 | 14.95 | 0.0490 |
| Evidential Backing | 4.79 | 4.17 | 0.0346 |
| Super claim | 1.08 | 0.71 | 0.0426 |
| Rebuttal | 24.24 | 22.91 | 0.0853 |
| Warrant | 0.30 | 0.00 | 0.0087 |
| Self-repetition | 5.62 | 5.57 | 0.0272 |
| Team-repetition | 11.53 | 11.58 | 0.0326 |
| Reciting | 9.12 | 9.22 | 0.0236 |

## 5 Conclusion

This study introduces an argument annotation model applied to a corpus of Arabic competitive debates. It uses a hybrid argumentation analysis model combining Aristotle's and Toulmin's models with some extra labels added as a response to insights emerging from the annotation process of the dataset, as shown in Table 5. The study situated Toulmin's model within Aristotle's logos as it is the focus of Toulmin's model to delineate the logical structure of an argument but had to account for the other two appeals of ethos and pathos as these two were quite explicit in the data and could not have been ignored. The authors developed an annotation model and verified and tested it on a sizable and diverse corpus of Arabic debates. The initial findings of the annotation confirm previous research on Arabic argumentation. The model developed in this study will be

tested and verified further with more data from the same Arabic debating corpus as the annotation process progresses with the rest of the debates in the corpus.

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